

Software Requirements Specification (SRS) Document

Project Name: E-Restaurants Ordering System

1. Introduction

1.1 Purpose

The purpose of the **E-Restaurants Ordering System** is to provide an online platform for customers to browse restaurant menus, place orders, and make payments seamlessly. The system will also enable restaurant owners to manage menus, process orders, and track sales efficiently.

1.2 Scope

The system will have three major user roles: Customers, Restaurant Owners, and Administrators. The platform will include features like:

- User registration and authentication
- Menu browsing and item selection
- Order placement and real-time tracking
- Online payment integration
- Admin dashboard for restaurant management
- Notification system (email/SMS)
- Review and rating system
- Order history and reports generation

1.3 Definitions, Acronyms, and Abbreviations

- **POS:** Point of Sale
- **UAT:** User Acceptance Testing
- **API:** Application Programming Interface
- **UI/UX:** User Interface/User Experience

1.4 References

- IEEE SRS Template

- Online Restaurant Management System Guidelines

1.5 Overview

This document describes the functional and non-functional requirements of the **E-Restaurants Ordering System**, system features, and user roles.

2. Overall Description

2.1 Product Perspective

The system will be a web-based application that integrates with payment gateways, notification services, and a database to store order details.

2.2 Product Functions

- Secure user authentication and role-based access control
- Intuitive UI/UX for easy menu navigation and ordering
- Automated order processing for restaurants
- Payment gateway integration (PayPal, Stripe, etc.)
- Reports and analytics for sales tracking

2.3 User Characteristics

- **Customers:** Users placing food orders via the platform.
- **Restaurant Owners:** Business owners managing their restaurant's menu, orders, and earnings.
- **Administrators:** Manage overall system operations, security, and support.

2.4 Constraints

- The system must support concurrent users without performance issues.
- Must comply with data privacy and security regulations.
- Limited to specific geographic regions initially.

2.5 Assumptions and Dependencies

- Users have stable internet access.
- Restaurants manage their own menus and availability.
- The platform relies on third-party payment gateways and SMS/email services.

3. Specific Requirements

3.1 Functional Requirements

- **User Authentication:** Secure login and role-based access.
- **Menu Management:** Restaurants can add, update, and remove items.
- **Order Placement & Tracking:** Users can place, modify, and track orders.
- **Payment Processing:** Support for credit/debit cards, digital wallets.
- **Notifications:** Email/SMS alerts for order status updates.
- **Reviews & Ratings:** Customers can provide feedback.

3.2 Non-Functional Requirements

- **Scalability:** Support multiple restaurants and concurrent users.
- **Security:** Implement SSL encryption, secure API calls, and user authentication.
- **Usability:** User-friendly interface with responsive design.
- **Performance:** Quick response time (<3 sec per request).

4. External Interface Requirements

4.1 User Interfaces

- Web-based UI with a mobile-responsive design.
- Interactive dashboards for restaurant owners.
- Order tracking interface for customers.

4.2 Hardware and Software Requirements

- Web Server: Apache/Nginx
- Database: MySQL/MongoDB
- Backend: Node.js with Express
- Frontend: React/Angular

5. Other Considerations

- **Data Privacy Compliance:** GDPR, PCI DSS for secure transactions.
- **Future Enhancements:** AI-driven recommendations, chatbot support.

6. Appendices

- User flow diagrams
- API documentation (if applicable)

